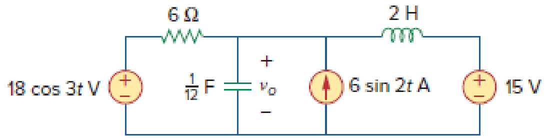
Chapter 10, Problem 46P

Problem

Solve for *vo*(*t*) in the circuit of Fig. 10.91 using the superposition principle.

Figure 10.91:



Step-by-step solution

Step 1 of 9

Refer to Figure 10.91 in the textbook.

The circuit is operating at three different frequencies, that is. $0\ \text{rad/s}$ (for DC source), $2\ \text{rad/s}$ and $3\ \text{rad/s}$.

Use the super position principle.

Assume that the total voltage due to three sources is,

$v_0 = v_1 + v_2 + v_3$ ...... (1)

Here,

v_1 is due to the $18\cos(3t)\,\text{V}$ voltage source,

v_2 is due to the $6\sin 2t\ \text{A}$ current source,

v_3 is due to the $15\ \text{V}$ dc voltage source.

Step 2 of 9

Find the output voltage v_1. Consider that the voltage source $18\cos(3t)\,\text{V}$ is acting alone and set both the sources $6\sin 2t\ \text{A}$ current source and $15\ \text{V}$ dc voltage source to zero and transform the circuit into frequency domain.

Thus,

$$18\cos(3t)\,\text{V} = 18\angle 0°\ \text{V}$$

Here, the angular frequency ω is 3 rad/s.

Determine the impedance of the inductor.

$$\begin{aligned} j\omega L &= j(3)(2) \\ &= j6\ \Omega \end{aligned}$$

Determine the impedance of the capacitor.

$$\begin{aligned} \frac{1}{j\omega C} &= \frac{1}{j(3)\left(\frac{1}{12}\right)} \\ &= -j4\ \Omega \end{aligned}$$

Step 3 of 9

Draw the frequency domain equivalent circuit with source $18\angle 0^\circ$ V as shown in Figure 1.

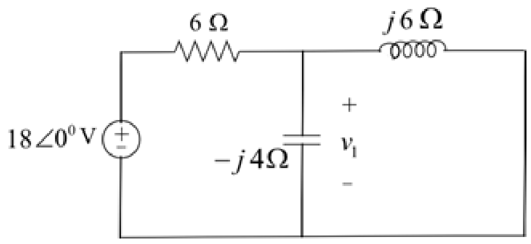


Figure 1

Step 4 of 9

Apply the Kirchhoff's current law at node, v_1.

$$\frac{v_1 - 18\angle 0^\circ}{6} + \frac{v_1}{-j4} + \frac{v_1}{j6} = 0$$

$$v_1\left(\frac{1}{6} + \frac{1}{-j4} + \frac{1}{j6}\right) = 3$$

$$v_1(0.167 + j0.083) = 3$$

$$\begin{aligned} v_1 &= \frac{3}{0.167 + j0.083} \\ &= 14.4 - j7.2 \\ &= 16.099\angle -26.56^\circ \text{ V} \end{aligned}$$

In the time domain,

$v_1 = 16.099\cos(3t - 26.56^\circ)$ V …… (2)

Step 5 of 9

Find the output voltage v_2. Consider that the current source $6\sin 2t$ A is acting alone and set both the sources $18\cos(3t)$ V voltage source and 15 V dc voltage source to zero and transform the circuit into frequency domain.

Thus,

$$\begin{aligned} 6\sin 2t \text{ A} &= 6\cos(2t - 90^\circ)\text{ A} \\ &= 6\angle -90^\circ \text{ A} \end{aligned}$$

Here, the angular frequency ω is 2 rad/s.

Determine the impedance of the inductor.

$$\begin{aligned} j\omega L &= j(2)(2) \\ &= j4\ \Omega \end{aligned}$$

Determine the impedance of the capacitor.

$$\begin{aligned} \frac{1}{j\omega C} &= \frac{1}{j(2)\left(\frac{1}{12}\right)} \\ &= -j6\ \Omega \end{aligned}$$

Step 6 of 9

Draw the frequency domain equivalent circuit as shown in Figure 2.

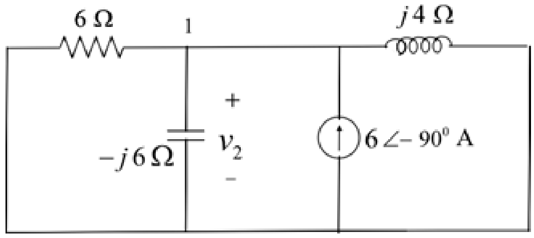


Figure 2

Step 7 of 9

Write the Kirchhoff's current law equation at node 1.

$$\frac{v_2}{6}+\frac{v_2}{-j6}+\frac{v_2}{j4}=6\angle-90^\circ$$
$$v_2\left(0.167+j0.167-j0.25\right)=6\angle-90^\circ$$
$$v_2\left(0.167-j0.083\right)=6\angle-90^\circ$$
$$v_2=\frac{6\angle-90^\circ}{0.167-j0.083}$$
$$v_2=\frac{6\angle-90^\circ}{0.186\angle-26.427^\circ}$$
$$=32.2\angle-63.43^\circ\text{ V}$$

In the time domain,

$v_2=32.2\cos\left(2t-63.43^\circ\right)\text{ V}$ ...... (3)

Step 8 of 9

Find the output voltage v_3. set all the sources to zero except the 15 V dc source.

In the steady state condition, the capacitor acts like an open circuit and inductor acts like a short circuit.

Draw the equivalent circuit as shown in Figure 3.

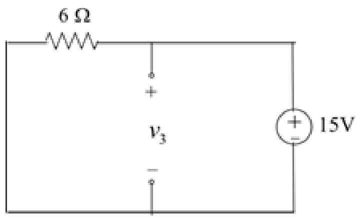


Figure 3

Step 9 of 9

From Figure 3,

$v_3=15\text{ V}$ ...... (4)

Recall equation (1).

$$v_0=v_1+v_2+v_3$$
$$=16.099\cos\left(3t-26.56^\circ\right)+32.2\cos\left(2t-63.43^\circ\right)+15\text{ V}$$

Therefore, the output voltage v_0 is

$$\boxed{16.099\cos\left(3t-26.56^\circ\right)+32.2\cos\left(2t-63.43^\circ\right)+15\text{ V}}.$$